

Graduation Project Proposal Form

1. Project Information

- **Project Title:** UK Train Ride Analytics & Insights Dashboard Using Power BI
- **Course/Track:** Microsoft Power BI Specialist
- **Team Members:**
 1. Shereen Talaat Saad
 2. Rimon Ezzat Habeeb
 3. Basma Atef Abd El_Ghafour
 4. Mona Mohammed Hussein

2. Project Overview

Objective:

The objective of this project is to leverage business intelligence techniques to analyze UK train ride data and uncover patterns that influence revenue generation, customer behavior, and operational reliability.

Using Microsoft Power BI, the project transforms complex datasets into dynamic dashboards that highlight key performance indicators (KPIs), including ticket sales trends, delay frequency, cancellation rates, and refund impact.

The ultimate goal is to provide data-driven recommendations that can enhance operational performance, optimize revenue streams, and support strategic planning within the transportation industry.

Scope of Work:

1- Data Preparation & Modeling

This project encompasses the end-to-end business intelligence process, starting from data preparation to insight generation and dashboard development using Microsoft Power BI.

The scope includes data cleaning and transformation using Power Query, ensuring data quality, handling missing values, correcting data types, and preparing the dataset for analysis.

A structured data model is designed to optimize performance, including the creation of relationships between tables and the implementation of calculated columns and DAX measures to compute key performance indicators (KPIs).

2- Exploratory Data Analysis

The project further covers exploratory data analysis to examine revenue performance, ticket sales distribution, passenger demand trends, delay patterns, cancellation rates, and refund behavior.

3- Dashboard Development & Insights Delivery

Interactive dashboards are developed to visualize findings through dynamic filters and drill-through capabilities, enabling stakeholders to monitor operational efficiency and identify areas for improvement.

Finally, the project delivers actionable insights and business recommendations based on data-driven analysis.

Expected Outcomes:

- **The expected outcomes of this project include delivering a comprehensive and interactive Power BI dashboard that provides clear visibility into key operational and financial performance indicators within the UK railway system.**
- **The project is expected to generate actionable insights regarding revenue trends, passenger demand behavior, peak travel periods, route performance, and service reliability. Additionally, it aims to identify patterns related to delays, cancellations, and refund requests, and assess their impact on overall business performance.**
- **By transforming raw transactional data into meaningful visual insights, the project will enable data-driven decision-making, support operational optimization, enhance revenue management strategies, and contribute to improving customer experience.**
- **Ultimately, the outcome is a professional business intelligence solution that demonstrates strong analytical capabilities and practical application of Microsoft Power BI tools and techniques.**

3. Problem Statement

- **Railway transportation systems generate large volumes of transactional and operational data daily. However, without proper analysis and visualization, this data remains underutilized and fails to provide meaningful insights for decision-makers.**
- **In the UK train ride dataset, key business challenges include understanding revenue fluctuations, identifying peak demand periods, evaluating service punctuality, and assessing the financial impact of delays, cancellations, and refund requests.**
- **The absence of a centralized analytical solution makes it difficult to monitor performance trends, detect operational inefficiencies, and optimize revenue strategies effectively.**
- **Therefore, there is a need to transform raw train ride data into an interactive and insightful business intelligence solution using Microsoft Power BI, enabling stakeholders to make informed, data-driven decisions and improve overall operational performance.**

4. Proposed Solution

Technologies Used:

- **Microsoft Power BI – For data visualization, dashboard development, and reporting**
- **Power Query – For data cleaning, transformation, and preparation**
- **DAX (Data Analysis Expressions) – For creating calculated columns, measures, and KPIs**
- **Excel / CSV – Source data handling and initial exploration**
- **Data Modeling – Star Schema**

System Architecture:-

- **1-Executive Dashboard: Management / Decision Makers**
- **2- Flow Analysis Dashboard :Planning / Strategic Operations Team**
- **3- Rail Performance Dashboard: Operations Team / Station Managers**
- **4- Sales Analysis Dashboard: Finance / Business Development Team**

5. Resources Needed

- **Hardware:**
Laptop.....

- **Software:**
 - **Microsoft Power BI Desktop** – Primary tool for data modeling, analysis, and dashboard development
 - **Microsoft Excel / CSV** – For initial data exploration and preparation
 - **Power Query (built-in Power BI)** – For data cleaning and transformation
 - **DAX (Data Analysis Expressions)** – For creating calculated columns, measures, and KPIs

6. Approval

- **Instructor/Advisor:**

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- **Signature:**

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